

CSE 604

Artificial Intelligence

Chapter 2: Intelligent Agents

Adapted from slides available in Russell & Norvig's textbook webpage

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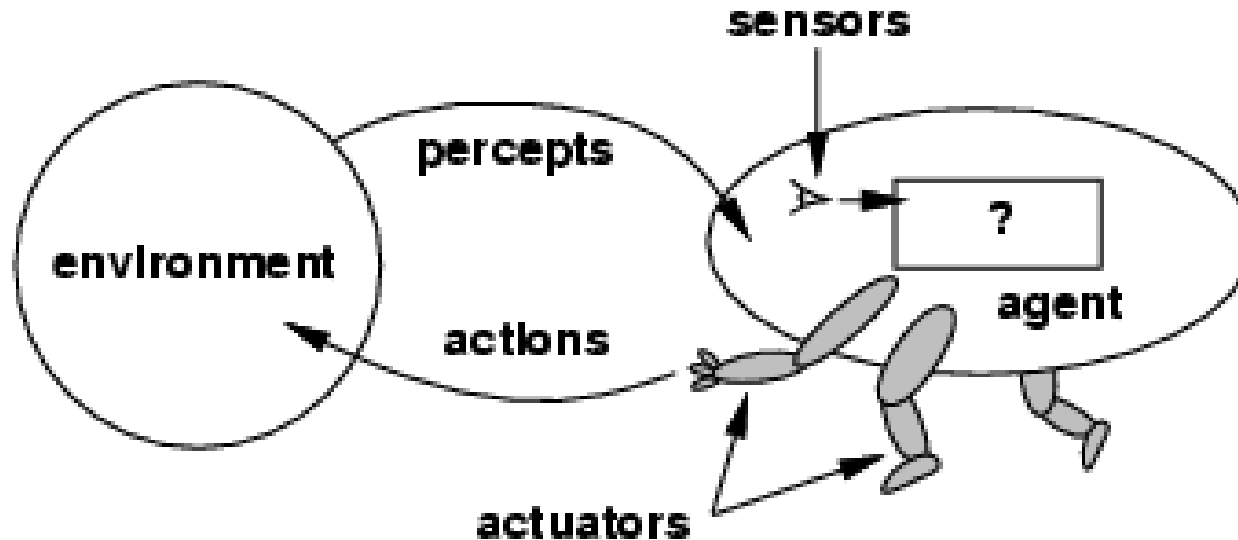


Outline

- Agents and environments
- Rationality
- PEAS (Performance measure, Environment, Actuators, Sensors)
- Environment types
- Agent types

Agents

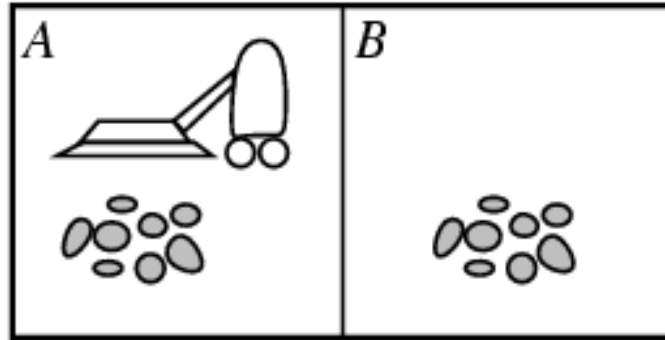
- An **agent** is anything that can be viewed as **perceiving** its environment through sensors and **acting** upon that environment through actuators



Examples of Agents

- **Human agent:** eyes, ears, and other organs for sensors; hands, legs, mouth, and other body parts for actuators
- **Robotic agent:** cameras and infrared range finders for sensors; various motors for actuators
- **Software agent:** receives keystrokes, file contents, network packets as sensory inputs; acts by displaying on screen, writing files etc.

Vacuum-cleaner world



- **Percepts:** location and contents, e.g., [A, Dirty]
- **Actions:** *Left*, *Right*, *Suck*, *NoOp*

Rational Agent

- A rational agent chooses whichever action maximizes the expected value of the performance measure given the percept sequence to date
- Rational $\neq$ omniscient
 - percepts may not supply all relevant information
- Rational $\neq$ clairvoyant
 - action outcomes may not be as expected
- Hence, rational $\neq$ successful
- Rational $\Rightarrow$ exploration, learning, autonomy

Rational agents

- An agent should strive to "**do the right thing**", based on what it can perceive and the actions it can perform.
- **Performance measure**: An objective criterion for success of an agent's behavior
- E.g., performance measure of a vacuum-cleaner agent could be :
 - amount of dirt cleaned up
 - amount of time taken
 - amount of electricity consumed

PEAS

- Specifying the task environment:
 - **P**erformance measure
 - **E**nvironment
 - **A**ctuators
 - **S**ensors

PEAS

- Agent: **Part-picking robot**
 - **Performance measure:** % of parts in correct bins
 - **Environment:** Conveyor belt, parts, bins
 - **Actuators:** Jointed arm and hand
 - **Sensors:** Camera, joint angle sensors



PEAS

- Agent: **Automated car**
 - **Performance measure:** Safe, fast, legal, comfortable trip
 - **Environment:** Roads, other traffic, pedestrians
 - **Actuators:** Steering wheel, accelerator, brake
 - **Sensors:** Camera, GPS, Speedometer, engine sensor

Environment types

- Fully observable vs. partially observable
- Single agent vs. multiagent
- Deterministic vs. stochastic
- Episodic vs. sequential
- Static vs dynamic
- Discrete vs continuous

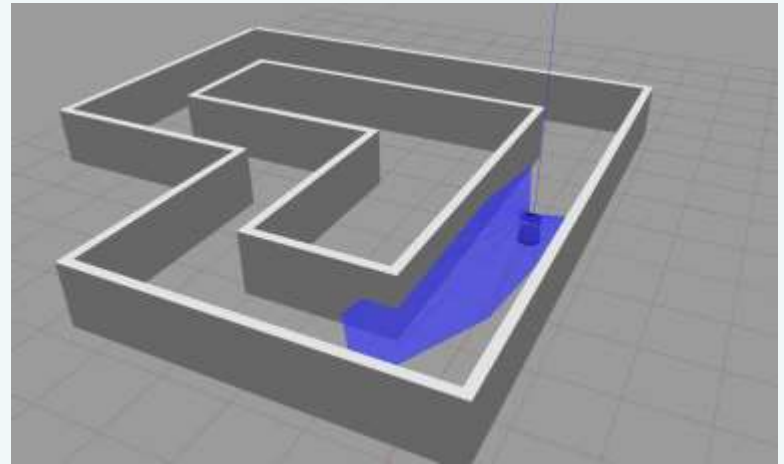
Fully Observable

Agent can observe (see/hear/perceive)
all relevant information from the
environment



Partially Observable

Agent can observe only partial
information from the environment



Single Agent

Our agent is the only intelligent agent in the environment



Multiagent

There are multiple intelligent agents which can be either **cooperative** or **competitive**



Deterministic

Agent can **fully determine** the outcome of its action (next step, not necessarily the full task)



Stochastic

Agent is **uncertain** of the outcome of its action



Episodic

Agent's actions are completely **independent** of each other, not linked to past or future actions



Sequential

Agent's actions are **dependent** on its past/future actions. The actions form a **sequence**.



Static

While the agent is in the process of taking it's action, the environment **doesn't change**



Dynamic

The environment is **constantly changing** even when the agent is taking an action



Discrete

Agent's task can be broken down into **discrete set of actions** (you can make a list of agent's actions $A_1, A_2, \dots, A_n$)



Continuous

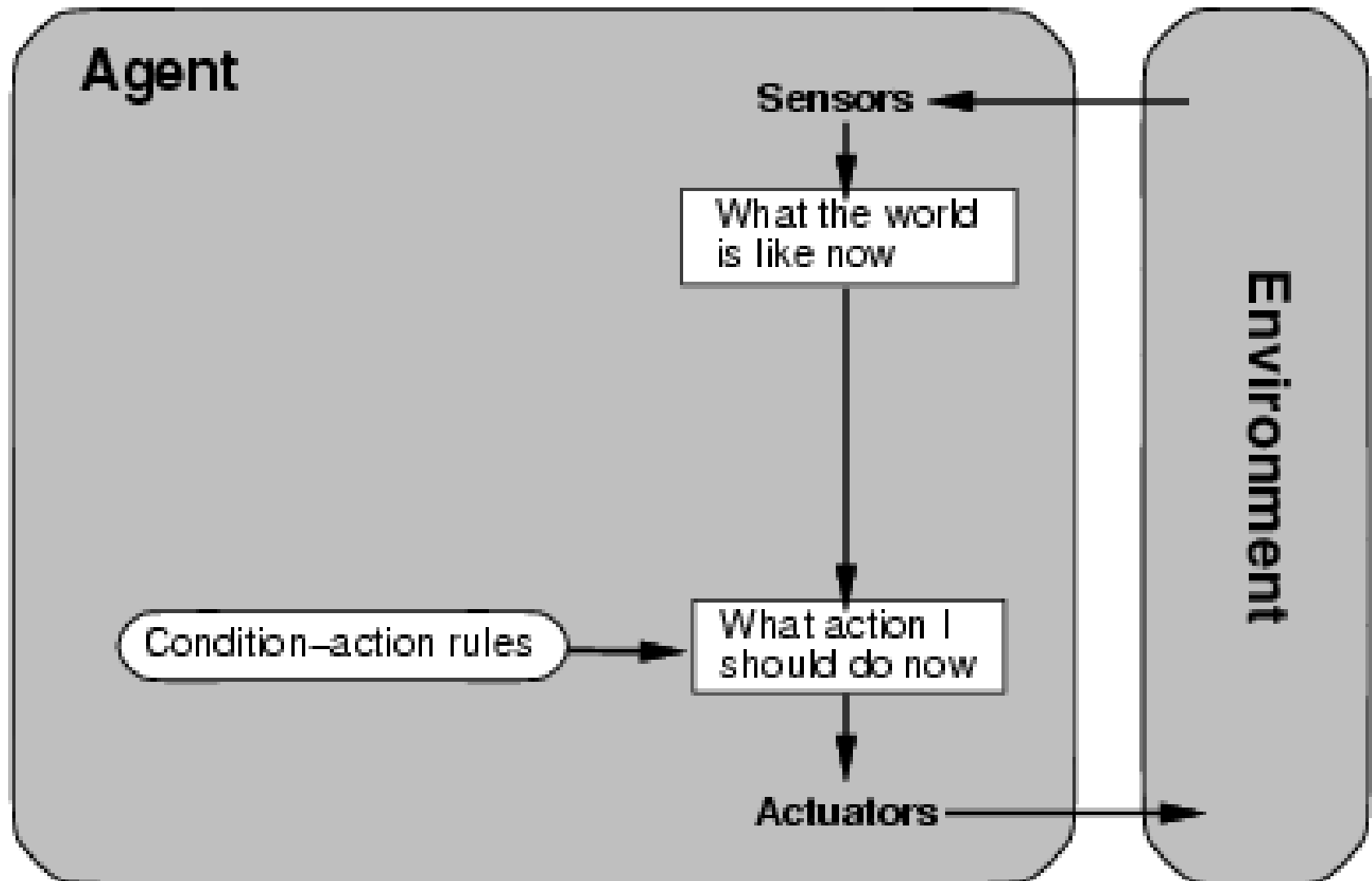
Actions are happening **continuously** and can not be listed, i.e., you cannot say where one action ends and the other begins



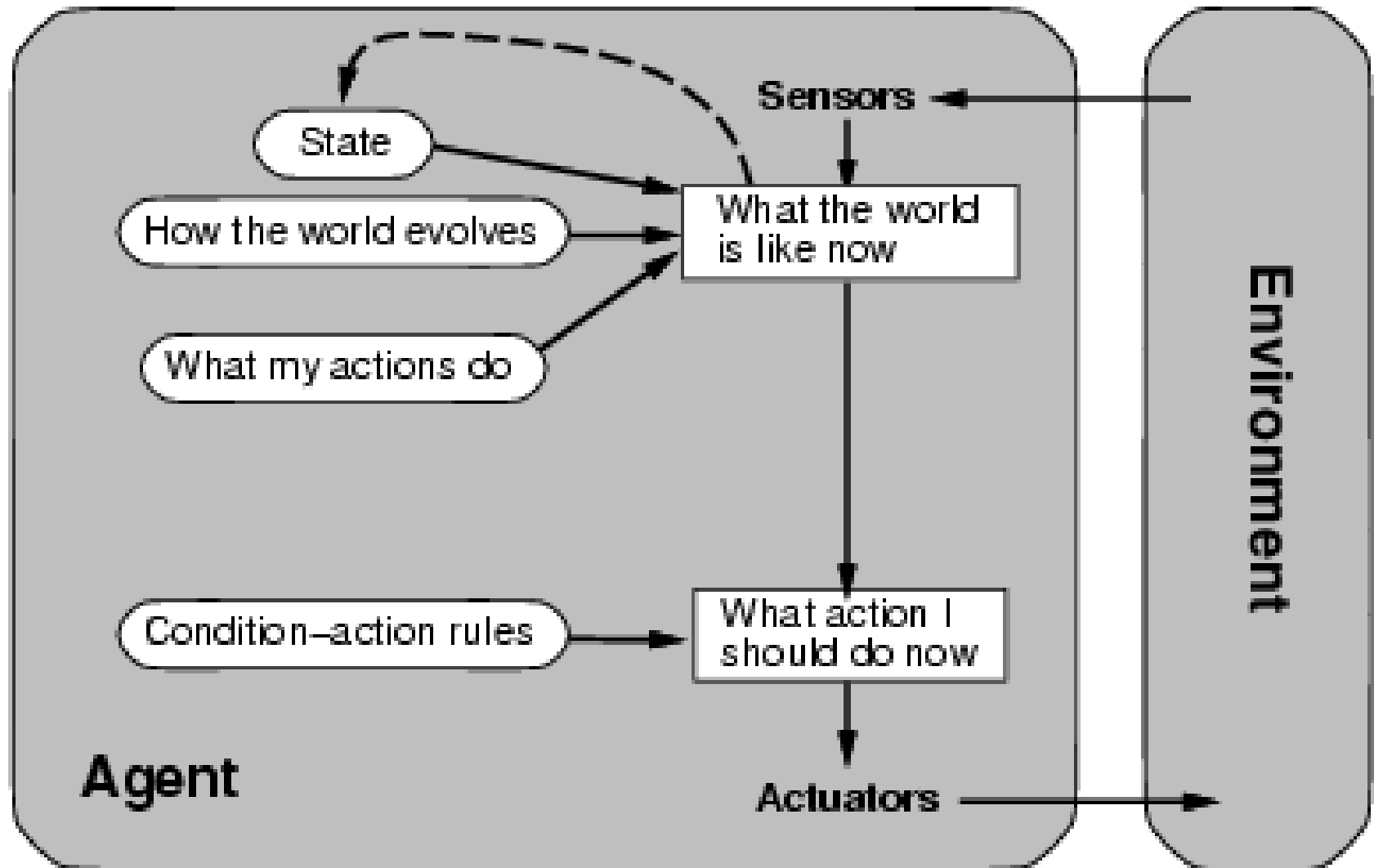
Agent types

- Four basic types in order of increasing generality:
 - Simple reflex agents
 - Model-based reflex agents
 - Goal-based agents
 - Utility-based agents

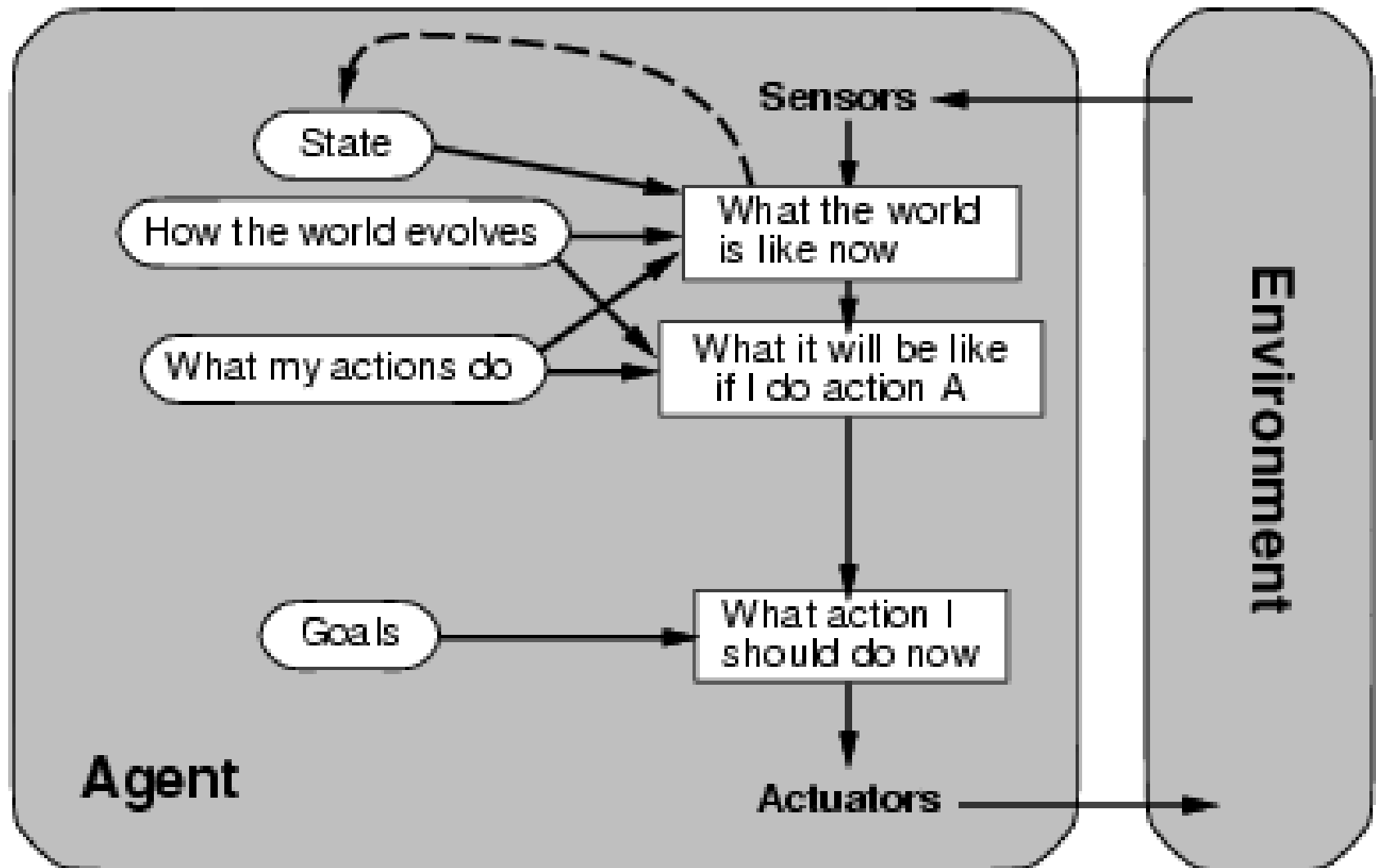
Simple reflex agents



Model-based reflex agents



Goal-based agents



Utility-based agents

